Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Medical · Class 11

Subject: Biology

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Cell Structure: Parts of a Plant and Animal Cell

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Unit 1: Cell Structure: Parts of a Plant and Animal Cell

Learning Objectives

• Students will name the main parts of plant and animal cells and state one function of each.

• Students will compare plant and animal cells, noting the cell wall, chloroplasts and large vacuole in plants.

• Students will draw and label a simple plant cell and a simple animal cell.

Key Concepts

• The cell is the basic unit of life; all living things are made of one or more cells.

• The cell membrane controls what enters and leaves the cell, while the cytoplasm is the jelly-like material where cell activities take place.

• The nucleus controls the cell and carries the genetic material that guides its activities.

• Mitochondria release energy from food, and ribosomes make proteins for the cell.

• Plant cells also have a rigid cell wall for support, chloroplasts for photosynthesis, and a large central vacuole for storage; animal cells lack these three structures.

Teaching Methodology (Step by Step)

1. Ask students what a brick is to a building, and link the idea to the cell as the unit of life.
2. Draw a labelled animal cell and a labelled plant cell side by side on the board.
3. Explain each part in turn, asking students to state its function before confirming it.
4. Students complete a comparison table of plant and animal cells from memory.
5. Students draw and label both cells, checking each other's labels.

Classroom Activities

• Cell city: groups match each cell part to a part of a city, such as the nucleus to the town hall.

• Diagram race: pairs label a blank cell diagram against the clock.

• Microscope pictures: students study printed cell photographs and pick out the nucleus and cell wall.

Assessment Questions

1. Name the part of the cell that controls its activities.
2. Give two structures found in plant cells but not in animal cells.
3. What is the function of mitochondria?

Homework

• Draw and label a plant cell and an animal cell, showing at least six parts in each.

• Explain in four lines why a plant cell needs a cell wall but an animal cell does not.

Time Allocation

40 minutes